Mainak Roy

AI/ML Engineer · Data Scientist

+91-6294131952 | mainakroy2003@gmail.com | mainak.roy23@st.niituniversity.in | LinkedIn | GitHub | Portfolio | Kolkata, West Bengal

OBJECTIVE

Data Science Undergraduate and **AI/ML Engineer** with hands-on experience in **Agentic AI**, **RAG pipelines**, and **Python backend development**, seeking to build reliable, context-aware AI systems through **LLM engineering**, **context engineering**, and **evaluation-driven development**.

EDUCATION

NIIT University — Neemrana, Rajasthan

Bachelor of Technology in Computer Science and Engineering, Data Science Specialization | **CGPA: 7.55** — *07/2023 – 08/2027*

- Relevant Coursework: Predictive Modeling for Data Science, Numerical Methods for Data Science, Inferential Statistics for Data Science, Big Data Concepts, Artificial Intelligence, Object-Oriented Programming

TECHNICAL SKILLS

Data Science: Pandas, NumPy, EDA, Data Cleaning, Feature Engineering, Statistical Analysis, Predictive Analytics
ML Libraries: Scikit-Learn, TensorFlow, OpenCV, Matplotlib, Seaborn
Deep Learning: NLP, CNN, U-Net, Transfer Learning, Neural Networks
Generative & Agentic AI: AI Agents, Multi-Agent Workflows, LangChain, LangGraph, LLM APIs, RAG, MCP
Programming: Python, Java, SQL
APIs & Web: REST APIs, JSON, Express.js, NestJS, FastAPI, Struts, JSP, Servlet, Tomcat, Microservices, JDBC, DAO Pattern
Frontend Development: Next.js
Database: MySQL, MongoDB, PostgreSQL, Snowflake
Cloud, DevOps: AWS (EC2, S3, Lambda), HDFS, Docker, Jenkins, CI/CD Pipelines
Version Control: Git, GitHub
Data Visualization: Power BI, Excel

EXPERIENCE

Software Development Engineer Intern — 06/2026 – 08/2026

G.S. & Co. Pvt. Ltd. (GSPL) — Delhi

Engineered and optimized robust backend modules for an enterprise web application utilizing the Apache Struts Model-View-Controller (MVC) framework, Servlets, and JSP technology. Designed and refactored Data Access Objects (DAOs) using Java Database Connectivity (JDBC) to interface seamlessly with a PostgreSQL database, authoring complex, highly performant SQL queries. Strengths included driving end-to-end full-stack integration, debugging data-type mismatches between front-end UI components and back-end controllers, and implementing role-based access control (RBAC) logic in Action classes to secure key workflows and optimize overall application responsiveness.

PROJECTS

TripMate AI: Multi-Agent Travel Planning System | *Python, FastAPI, LangGraph, LangChain, MCP, PostgreSQL, Groq* — Live Demo

- Built a LangGraph supervisor workflow that routes travel requests to specialized flight, hotel, weather, budget, and itinerary agents, integrating external data through MCP
- Implemented PostgreSQL-backed workflow checkpoints and human-in-the-loop review, allowing users to approve or request revisions to draft itineraries
- Developed a FastAPI web interface with request validation, sanitized Markdown rendering, and PDF export; containerized the application with Docker

MedMes CareFlow: Multi-Agent Healthcare AI Orchestration System | *Python, LangChain, FastAPI, RAG* — Live Demo

- Designed a role-based multi-agent AI system with 4 specialized clinical agents (Patient Navigator, Care Coordinator, Quality Analyst, AI Nurse) orchestrated by a master routing agent using LangChain and Google Gemini 1.5 Flash
- Built a RAG pipeline grounded on 15 clinical guidelines (ADA, AHA, KDIGO, ACC/AHA) enabling evidence-based decisions across FHIR-compatible synthetic patient data covering lab results, vitals, medications, and care gaps
- Implemented a real-time evaluation framework tracking routing confidence, task completion rate, and response latency, achieving over 90% routing accuracy aligned with MLOps standards for healthcare AI safety

ZonePulse: Uber/Ola Ride Demand Prediction System | *Python, XGBoost, Snowflake, Streamlit, SQL* — GitHub

- Built a ride demand forecasting pipeline processing 4.7 million records from a Snowflake cloud data warehouse, reducing feature extraction time by 40% through optimized SQL query design
- Trained XGBoost and Random Forest models on spatiotemporal trip data, achieving an R-squared of 0.78 (78% variance explained) in ride demand prediction to support surge pricing and driver allocation decisions
- Deployed interactive dashboards with geospatial demand heatmaps, enabling operations teams to identify peak demand windows (5 PM) and high-density zones across 12+ city districts

NeuroSeg: Brain Tumor Detection System | *Python, TensorFlow, U-Net, VGG19, Grad-CAM, Docker, Jenkins, CI/CD* — GitHub

- Engineered a U-Net architecture for MRI brain tumor segmentation, achieving 97.2% accuracy, and fine-tuned a VGG19 transfer learning model that reached 99.1% classification accuracy, outperforming the ANN baseline (82.4%) and standard CNN (94.8%) across all evaluation metrics
- Integrated Grad-CAM heatmaps for clinical explainability and automated deployment via a CI/CD pipeline (GitHub to Jenkins to Docker to Docker Hub), reducing manual release effort by 100% through zero-touch delivery

PUBLICATIONS

A Comparative Experimental Study of Maze-Solving Algorithms in Known and Unknown Environments — GitHub — 2026
— ICICV 2026 (Accepted)

CERTIFICATIONS & LEADERSHIP

ISRO Bharatiya Antariksh Hackathon 2025: AI-Based Air Pollution Monitoring and Forecasting System — GitHub — 2025

Built RAG Chatbot with Python — LetsUpgrade Certification, May 2025

Club Coordinator, Ramanujan Mathematics Club, NIIT University — 03/2025 – 04/2026